Shouren Wang

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LinkedIn | GitHub | Google Scholar

Education

Case Western Reserve University — Cleveland, OH
Ph.D. in Computer Science | *GPA: 3.83/4.0* — *Aug. 2024 – Present*

New York University — Brooklyn, NY
M.S. in Computer Engineering | *GPA: 3.93/4.0* — *Sep. 2021 – May 2023*

Hunan University — Changsha, China
Undergraduate in Software Engineering — *Sep. 2015 – Jun. 2017*
Undergraduate and B.E. in Digital Media Technology | *GPA: 83.16/100.0* — *Jun. 2017 – Jun. 2019*
(A specialization of Software Engineering focusing on visual software/algorithm development)

Experience

Graduate Research Assistant — Aug. 2024 – Present
TAPP AI Lab, Case Western Reserve University — *Cleveland, OH*
- Conduct research on large language model reasoning, post-training, model architecture, and AI agent systems.
- Advisors: Xiaotian Han and Vipin Chaudhary.

Teaching Assistant — Sep. 2024 – May 2026
Case Western Reserve University — *Cleveland, OH*
- Teaching assistant for *Large Language Models* and *Database Systems*: graded assignments and held office hours.
- Professors: Xiaotian Han (Large Language Models); Yinghui Wu (Database Systems).

Research Intern — Aug. 2023 – Jul. 2024
NYU Game Innovation Lab — *Brooklyn, NY*
- Explored the research and methods for creative AI for video games.
- Developed enjoyment-focused game-playing agents for Street Fighter II via an extended PPO model and self-play.
- Advisor: Julian Togelius, Professor, New York University.

Research and Development Engineer — Sep. 2019 – Sep. 2020
AsiaInfo Technology — *Nanjing, China*
- Tested the CTDI project and improved its performance as a member of the QA team.
- Manager: Ted Wang.

Projects

How to Loop MoE: Flatten the Experts, Untie the Attention — Jul. 2026 – Present
[arXiv] | [GitHub] | *LLMs, Model Architecture, Mixture-of-Experts, Looped Model*
- Proposed **Foil**, a looped MoE architecture that flattens expert layers, loops them more often, and gives each pass its own attention parameters while holding expert parameters and expert compute per token fixed.
- Validated Foil with 20B-token pretraining and 100B-token continued training; every Foil model outperformed the unflattened baseline at 20B tokens, and the most flattened model finished 0.012 nat lower at 100B tokens with downstream accuracy on par or better.
- Characterized expert utilization through coverage, load balance, and routing confidence, showing that wider expert layers and additional passes amplify one another and that routing-confidence peaks help diagnose diminishing returns from looping.

Agent Team Work Zone — Apr. 2026 – Present
[arXiv] | [GitHub] | [npm] | *LLM Agents, Claude Code, Multi-Agent Workflow*
- Built and maintain an open-source persistence layer for Claude Code's Agent Teams, whose teammates vanish on session restart with no native recovery.
- Designed per-agent workstations (role, working context, checkpoints, commitments) and a team registry, so one command rebuilds an interrupted team from disk.
- Shipped 13 slash-command skills and hook-driven auto-checkpointing across four public releases, maintaining parallel editions through a breaking Claude Code API change.
- Use the system end-to-end in its own development workflow: the agent team that builds it also runs on it.

Path-Lock Expert: Efficient Hybrid Thinking Architecture Jan. 2026 – Jul. 2026
[arXiv] | [GitHub] | *LLMs, Reasoning*

- *Accepted at COLM 2026.*
- Developed a parameter-efficient architecture approach to improve hybrid-thinking *mode controllability* (thinking vs. fast) without compromising overall task performance. Built an efficient training pipeline with curated data.
- Experiments on reasoning benchmarks (AIME24, MATH500, GPQA-Diamond) indicate improved mode separation/controllability with competitive accuracy, and greatly improved accuracy for no-think mode.

Demystifying Hybrid Thinking Sep. 2025 – Feb. 2026
[arXiv] | [GitHub] | *LLMs, Reasoning, Post-Training*

- *Accepted at EMNLP 2026 Main.*
- Systematically analyzed the controllability of hybrid-thinking LLMs (think vs. no-think), showing incomplete mode separation and quantifying "reasoning leakage" via output length and reflection-token counts across multiple evaluations (MATH500/AIME24/GPQA/MMLU-STEM).
- Conducted controlled SFT ablations on data scale (20k–140k), pairing strategy, think:no-think ratio, and two-phase training; proposed a practical training recipe that preserves accuracy while cutting no-think verbosity (1085→585 tokens on MATH500) and reflection-token frequency (5917→522).

Mid-Think: Token-Level Reasoning Budget Control Nov. 2025 – Jan. 2026
[arXiv] | [GitHub] | *LLMs, Reasoning, RL*

- *Accepted at ACL 2026 Findings.*
- Identified via attention analysis that hybrid-thinking mode switching is dominated by a few *token-level triggers*—"Okay" activates reasoning while `</think>\n\n` suppresses it—rather than high-level Think/No-think instructions.
- Proposed **Mid-Think**, a training-free prompting format combining both triggers to achieve Pareto-optimal accuracy–length trade-offs (MATH500: 92.1% acc. at 2589 tokens vs. 94.4% at 4904), outperforming fixed-token and prompt-based baselines across Qwen3-4B/8B/14B/32B.
- Applied Mid-Think to GRPO-based RL after SFT, cutting training time ∼15% (54h→46h) while improving Qwen3-8B on AIME (69.8%→72.4%) and GPQA (58.5%→61.1%).

Publications

*Equal contribution. †Corresponding author(s).

[1] **Path-Lock Expert: Separating Reasoning Mode in Hybrid Thinking via Architecture-Level Separation**
COLM 2026

- Authors: **Shouren Wang***, Wang Yang*, Chuang Ma, Debargha Ganguly, Vikash Singh, Chaoda Song, Xinpeng Li, Xianxuan Long, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[2] **Demystifying Hybrid Thinking: Can LLMs Truly Switch Between Think and No-Think?**
EMNLP 2026 Main

- Authors: **Shouren Wang***, Wang Yang*, Xianxuan Long, Qifan Wang, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[3] **Mid-Think: Training-Free Intermediate-Budget Reasoning via Token-Level Triggers**
ACL 2026 Findings

- Authors: Wang Yang*, **Shouren Wang***, Debargha Ganguly, Xinpeng Li, Chaoda Song, Vikash Singh, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[4] **A Survey on Efficient Protein Language Models**
TMLR 2026

- Authors: **Shouren Wang**, Debargha Ganguly, Vinooth Rao Kulkarni, Wang Yang, Zhuoran Qiao, Daniel Blankenberg, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[5] **How to Loop MoE: Flatten the Experts, Untie the Attention**
Under Review

- Authors: **Shouren Wang***, Chuang Ma*, Mohsen Hariri, Debargha Ganguly, Wang Yang, Xiaoqing Tong, Qianying Liu, Xiaotian Han†, Vipin Chaudhary†
- Links: [arXiv], [GitHub]

[6] **Agent Team Work Zone: An Automated, Persistent Workspace for Long-Lived Claude Code Agent Teams**
Preprint, Open Source, npm Package

- Authors: **Shouren Wang**
- Links: [arXiv], [GitHub], [npm]

[7] **Test-Time Scaling in Reasoning LLMs: Inference Regimes, Evaluation, and Reproducibility**
Under Review

- Authors: Mohsen Hariri, Weicong Chen, Nahal Shahini, Vikash Singh, Kai Ye, Amirhossein Samandar, Debargha Ganguly, Sreehari Sankar, Yanyan Zhang, **Shouren Wang**, Jerry Peng, Biyao Zhang, Michael Hinczewski, Vipin Chaudhary
- Links: [arXiv]

[8] **When Domains Interact: Asymmetric and Order-Sensitive Cross-Domain Effects in Reinforcement Learning for Reasoning**
Under Review

- Authors: Wang Yang, **Shouren Wang**, Chaoda Song, Chuang Ma, Xinpeng Li, Nengbo Wang, Kaixiong Zhou, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[9] **A Survey on Agent Skills for LLMs: A Lifecycle Perspective from Construction to Ecosystems**
Under Review

- Authors: Wang Yang, Chaoda Song, Xinpeng Li, **Shouren Wang**, Nengbo Wang, Yanyan Zhang, Chuang Ma, Debargha Ganguly, Vikash Singh, Shuai Xu, Jing Ma, Yu Yin, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[10] **AgentCE-Bench: Agent Configurable Evaluation with Scalable Horizons and Controllable Difficulty under Lightweight Environments**
Under Review

- Authors: Wang Yang, Chaoda Song, Xinpeng Li, Debargha Ganguly, Chuang Ma, **Shouren Wang**, Zhihao Dou, Yuli Zhou, Vipin Chaudhary†, Xiaotian Han†
- Links: [Paper], [Repository]

[11] **MemTrace: Probing What Final Accuracy Misses in Long-Term Memory**
Under Review

- Authors: Xianxuan Long, Zhikai Chen, Shenglai Zeng, **Shouren Wang**, Kai Guo, Jiliang Tang†
- Links: [Paper]

[12] **Reliability-Gated Source Anchoring for Continual Test-Time Adaptation**
Under Review

- Authors: Vikash Singh, Debargha Ganguly, Weicong Chen, Sabyasachi Sahoo, Sreehari Sankar, Biyao Zhang, Mohsen Hariri, **Shouren Wang**, Osama Zafar, Christian Gagné, Vipin Chaudhary†
- Links: [Paper]

[13] **CausalGuard: Conformal Inference under Graph Uncertainty**
Under Review

- Authors: Vikash Singh, Weicong Chen, Debargha Ganguly, Yanyan Zhang, Nengbo Wang, Sreehari Sankar, Mohsen Hariri, Alexander Nemecek, Chaoda Song, **Shouren Wang**, Biyao Zhang, Van Yang, Erman Ayday, Jing Ma, Vipin Chaudhary†
- Links: [Paper]

Academic Service

Reviewer: Conference on Language Modeling (COLM) 2026
Reviewer: International Conference on Learning Representations (ICLR) 2027 (ongoing)

Technical Skills

Languages: Python, Java, MATLAB
Frameworks: PyTorch, Transformers, vLLM, Llama-Factory, Stable-Baselines3, OpenAI Gym, Scikit-Learn
Developer Tools: Git, Slurm, Anaconda, Docker, Singularity, Linux, VS Code
AI Development Tools: Claude Code, OpenClaw, Codex, Cursor

Relevant Graduate-Level Coursework

- **LLMs & Generative Models**: Large Language Models [CWRU, CSDS 600], Deep Generative Models [CWRU, CSDS 570]
- **Machine Learning**: Advanced Machine Learning [NYU, ECE-GY 7143], Machine Learning [NYU, ECE-GY 6143], Causality and Machine Learning [CWRU, CSDS 452], Data Mining [CWRU, CSDS 435]
- **Core CS & Mathematics**: Data Structures and Algorithms [NYU, ECE-GY 9343], Computing Systems Architecture [NYU, ECE-GY 6913], Probability and Stochastic Processes [NYU, ECE-GY 6303]